Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/27/2023
PI Name:	Alwood, Joshua Ph.D.		
Project Title:	Integrated CNS Assessment in Rodent Models of Altered Gravity and Irradiation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
PI Email:	joshua.s.alwood@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1490
Organization Name:	NASA Ames Research Center		
PI Address 1:	MS 261		
PI Address 2:	Building N261, Rm 113		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	03/01/2023	**End Date:**	02/28/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Desai, Rajeev Ph.D. (McLean Hospital Corp) Kangas, Brian Ph.D. (McLean Hospital Corp) Ronca, April Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To better understand gravity's role in life's processes, gravitational acceleration – from microgravity to hypergravity – is worthy of study in space-based and ground-based experiments across organisms. Additionally, little is known about potential interactions across the gravity continuum with exposure to ionizing radiation. In this proposed project, we aim to study the effects of gravity as a continuum, from simulated weightlessness with hindlimb unloading or from hypergravity, in combination with chronic low-dose irradiation, to assess the time course of changes in cognitive processes and associated neurotransmitter levels and circulatory biomarkers. Rats will be treated with ground analogs of spaceflight: either hindlimb unloading combined with irradiation in Aim 1, or hypergravity at 2g combined with irradiation in Aim 2. Hindlimb unloading will be achieved using custom plexiglass cages at NASA Ames Research Center (ARC). Hypergravity treatment through centrifugation will take place at NASA ARC. Low-dose irradiation will use 137Cs gamma-ray exposure. Cognitive assessment will use touchscreen and non-touchscreen. In vivo microdialysis will be performed to collect fluid for neurotransmitter assessment in key brain regions. Blood and hippocampus will be collected and assessed for complementary biomarkers. This project has the potential to advance both the state of knowledge and the state of technology to be useful in future space missions. Specifically, this project will further mechanistic underpinnings that can be associated with inflammation and oxidative stress in both the brain and circulation to determine mechanistic and translatable biomarkers integrated with gold-standard cognitive/behavioral assessment leading to peer-reviewed publications and publicly archivable data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Assessment of various neurocognitive domains that can be meaningfully linked to changes in brain neurotransmitters and circulatory biomarkers in the brain/peripheral system as critical areas of research to uncover the risk to cognitive-sensorimotor function. The proper functioning of these neurological processes is integral for maintaining astronaut health, cognitive performance, and overall mission outcomes.
Task Progress:	We are in the first year of a three-year project with our team at NASA Ames Research Center and McLean Hospital (Belmont, MA). In the first year of the project, we have conducted two experiments (one for males, one for females) using centrifugation at 2g for 28 days and gamma irradiation, which addresses Aim 2 of the project. We have reported our preliminary results at the 2023 American Society for Gravitational and Space Research (ASGSR) conference. In Year 2, we aim to complete 1 additional centrifugation experiment using male rats and to begin hindlimb unloading pilot studies with female rats. Additionally, we plan to continue our analysis of the first two experiments.
Bibliography Type:	Description: (Last Updated: 05/25/2026)
Abstracts for Journals and Proceedings	Tsekov S, Puukila S, Tabares Ruiz S, Mhatre S, Ronca AE, Kangas BD, Desai RI, Alwood JS. "Integrated responses to 2G centrifugation and irradiation on rat CNS." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Alwood JS, Ronca AE, Desai RI, Kangas BD, Puukila S, Tabares Ruiz S, Tsekov S. "Integrated CNS assessment in rodent models of altered gravity and irradiation." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Abstracts for Journals and Proceedings	Nishigaya D, Sanders L, Alwood J, Christenson LK, Costes S. "Histological and spatial transcriptomic sample preparation from spaceflight stressor exposed rat ovaries." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023